3.9 Operations With Absolute Value in the Real World

Lesson Introduction

Benchmark	MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute values.	Learning Target	• I can perform operations that involve the absolute values of two numbers.

	Building On	Working Toward
Benchmark Coherence	N/A	M.A.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers, including grouping symbols, whole-number exponents, and absolute value.

Essential Questions	• How do you perform operations with absolute values in real-world contexts? • How can number lines be used to create mathematical statements and perform operations with absolute values?
Lesson Narrative	Students model various real-world situations with absolute value expressions. Students perform operations with absolute values to solve problems in context.

Component Summary

3.9.1 Warm-Up (~5 min.)	Analyze a sale sign for an error *(SE p. 123)*	3.9.4 Try It (5 min.)	Practice writing and evaluating absolute value expressions in contexts *(SE p. 126)*
3.9.2 Collaboration (5-10 min.)	Interpret absolute values in the context of profit *(SE p. 123)*	3.9.5 Collaboration (10 min.)	Collaboratively write and evaluate absolute value expressions in contexts *(SE p. 126)*
3.9.3 Guided Instruction (10-15 min.)	Learn to write and evaluate absolute value expressions to solve problems *(SE p. 124)*	3.9.6 Your Turn! (10 min.)	Additional practice writing and evaluating absolute value expressions in contexts *(SE p. 127)*
3.9.7 Wrap-Up (~5 min.)	Self-assessment of understanding of absolute value learning targets *(SE p. 128)*		

Resources

Glossary	Materials	Additional Preparation
Key Terms: • Profit Additional Terms: N/A	• (Optional) Chart paper & markers	• Consider preparing images or videos of the contexts in the lesson to help students understand the contexts, as needed.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>
• Students may struggle with interpreting the meanings of the words revenue, expenses, and profit.
• Students may struggle interpreting the written descriptions of real-world contexts to write absolute value expressions.
• Students may add unnecessary (and time-consuming) details to their number lines.
</td><td>
• Consider connecting the terms “gross income,” “expenses,” and “net income” for an individual to the terms “revenue,” “expenses,” and “profit” for a business.
• Students have not yet learned to perform operations with integers. Students should determine the absolute values first and then perform the operations.
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<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
In 3.9.1 Warm-Up, students are asked what they notice and wonder about an image of a sale sign. Students should discuss what the error in the sale sign is and why it is not mathematically correct.</td><td>MA.K12.MTR.7.1: Real-World Contexts
In 3.9.6 Your Turn!, students interpret real-world situations to write absolute value expressions. Consider incorporating elements of students’ personal experiences to enhance their learning and make meaning of absolute value in their own contexts.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider providing sentence stems or answer choices for questions 4 and 5 in 3.9.5 Collaboration. This will help support students who struggle expressing themselves in writing to demonstrate their understanding.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

3.9.1 Warm-Up: Math Fail

Component Overview: *Students analyze a sale sign for errors. Students explain what the error is and why it is a mathematical error.*

3.9.1 Warm-Up: Math Fail

1. What do you notice about the sign?

Answers will vary. The sale price is greater than the original price, so that's not a good sale. Also, the math is not correct because $69.98 - 49 = 20.98$.

Notice and Wonder
Ask students what they notice and wonder about an image of a sale sign. Students should discuss what the error in the sale sign is and why it is not mathematically correct.

Consider challenging students to find more Math Fails as homework or extra credit as they review for the end of the unit.

3.9.2 Collaboration: Absolute Value and Profit

Component Overview: *Students work collaboratively to interpret and compare absolute values in context. The context explored involves revenue, expenses, and profit of a lawn mowing business.*

3.9.2 Collaboration: Absolute Value and Profit

1. Landon starts a lawn-mowing business to earn money. All of his profits are being saved for college.

   Profit is the difference between the amount earned (revenue) and the amount spent.

   His father lends him a lawnmower to use for free, but he must pay for the gas and other supplies. For each lawn, his expenses are $7.30, and he earns an average of $40.00.

   a. In this situation, revenue is represented with a ○ positive ○ negative value and an expense is represented by a ○ positive ○ negative value.

   b. Write a value to represent each situation.

Landon's Finances	Value
Expenses per lawn	−$7.30
Revenue per lawn	$40.00

   c. Use an inequality symbol, < or >, to compare the absolute values of Landon's expenses to revenue for each lawn he mows.

   $|\$40.00| > |-\$7.30|$

   d. How does the inequality show that Landon will make a profit?

   Answers will vary. The revenue he brings in is greater than his expenses, so he will make a profit.

   e. To make a profit, total expenses of a business must be ○ greater than ○ less than total revenue. A business is not profitable when the total ○ expenses are ○ revenue is greater than the ○ expenses. ○ revenue.

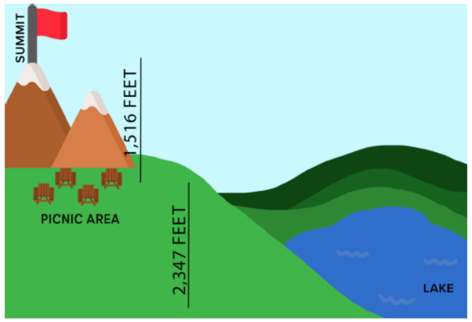
Consider having students create reference cards with vocabulary terms from the contexts explored in this lesson. Include definitions, examples, and visual models to improve student comprehension.

3.9.3 Guided Instruction: Operations With Absolute Value in the Real World

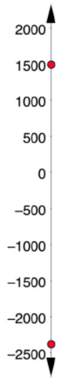
Component Overview: *Students learn to write and evaluate absolute value expressions in contexts. The contexts explored include elevation and finance.*

3.9.3 Guided Instruction: Operations with Absolute Value in the Real World

1. Javier and his family drove to a picnic area on a mountain. From the picnic area the family hiked to the mountain's summit, which was an elevation change of 1,516 feet (ft). They hiked back to the picnic area to eat lunch. After lunch, they hiked down to the lake, which was an elevation change of 2,347 ft.



   a. In the table, write the value that represents the position of each location if the value of the position of the picnic area is 0. Then, plot each value at the estimated location on the number line.



Location	Value
Picnic area	0
Mountain summit	1516
Lake	−2347

   b. Represent the distance of each location from the picnic area.

Location	Distance from Picnic Area
Mountain summit	1516 *ft.*
Lake	2347 *ft.*

- When calculating the distance between elevations, how do you know whether to add the absolute values or subtract them?
- How can the absolute values of a company's expenses and revenues be used to determine whether the company makes a profit?

c. What operation would you use with the representations in the table to find the distance from the lake to the mountain summit? *Addition*

d. What is the distance from the lake to the mountain summit? 3863 *feet*

e. What operation would you use to determine how much farther a hike to the lake is than a hike to the mountain summit? *Subtraction*

f. How much farther is a hike to the lake than a hike to the mountain summit? 831 *feet*

2. Chloe is starting a painting business. The supplies she purchased for the business cost $\$825$. She will earn a fee of $\$80$ for each room that she paints. Chloe wants to determine how many rooms she will need to paint before she starts making a profit.

a. ○ Fees Chloe earns / ○ Supplies Chloe buys increase her profit.

b. ○ Fees Chloe earns / ○ Supplies Chloe buys decrease her profit.

c. Write a value in the blank for each absolute value and a word in the remaining blank to complete each sentence.

To find the revenue, Chloe multiplies the number of rooms she paints by $|80|$.

To find the profit, Chloe subtracts $|-825|$ from the *revenue*.

d. In the table, complete the mathematical statement and then perform the operations to determine if the number of rooms painted would result in a profit.

Number of Rooms	Mathematical Statement	Profit (Yes or No)
6	$6\|80\| - \|-825\|$	*No*
10	$10\|80\| - \|-825\|$	*No*
13	$13\|80\| - \|-825\|$	*Yes*
16	$16\|80\| - \|-825\|$	*Yes*

Consider extending question 2 by asking early finishers the following:

- *Find the number of rooms Chloe needs to paint to break even.*
- *How would the mathematical statement change if Chloe earns a fee of $\$85$ dollars per room?*

3.9.4 Try It: Operations With Absolute Value in the Real World

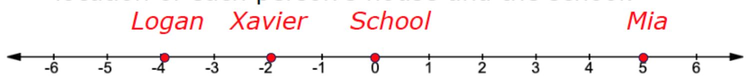
Component Overview: *Students practice writing and evaluating absolute value expressions in contexts. The context explored is locations of students' homes relative to their school. Students use a number line to aid them in answering the questions.*

3.9.4 Try It: Operations with Absolute Value in the Real World

Logan, Mia, and Xavier all live on the same street as their school, which runs from east to west.

- Logan lives 4 blocks to the west of the school.
- Mia lives 5 blocks to the east of the school.
- Xavier lives 2 blocks to the west of the school.

1. Use the number line to model the position of each house and the school along the street. Label the location of each person's house and the school.



2. Work with your partner to write the mathematical sentence using absolute values. Then, use the mathematical sentence to determine each answer.
   a. How far does Logan live from Mia?
   $|-4| + |5| = 9$ *blocks*

   b. How far does Mia live from Xavier?
   $|5| + |-2| = 7$ *blocks*

   c. How far does Xavier live from Logan?
   $|-4| - |-2| = 2$ *blocks*

   d. Was the same operation used to find each distance?
   No

Sample questions to consider to check for understanding as students work:

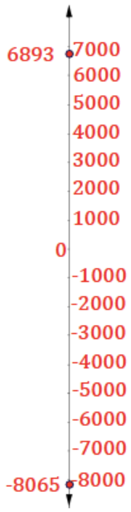
- *How did you determine where to plot the students' homes on the number line?*
- *How did you determine how far each student lives from school?*
- *How did you determine the distance between students' homes?*
- *How did you know whether to add or subtract the absolute values?*

3.9.5 Collaboration: Halfway Point

Component Overview: *Students work with a partner to determine the appropriate intervals for a number line used to model the elevations of Mount Ojos and the Atacama Trench. Students discuss how to determine the halfway point between the two locations and whether it is above or below sea level.*

3.9.5 Collaboration: Halfway Point

Mount Ojos del Salado is the highest mountain in Chile. Its peak is 6,893 meters above sea level. The lowest point of the Atacama Trench, just off the coast of Peru and Chile, is 8,065 meters below sea level.



Work with your partner to answer each question.

1. Determine how to label the number line so that you can approximate a location on the number line that represents Mount Ojos del Salado, Atacama Trench, and sea level.
   Answers will vary. Answer shown on number line.
2. Label the number line and place a point at the approximate location for Mount Ojos del Salado, Atacama Trench, and sea level.
   Answer shown on number line.
3. Write a mathematical sentence using absolute value to find the difference in elevations between Mount Ojos del Salado and the Atacama Trench.
   $|-8065| + |6893| = 14{,}958$
4. With your partner, discuss how the halfway point between the peak of Mount Ojos del Salado and the Atacama Trench can be found. Write down all of the strategies that were discussed.
   Answers will vary. Divide the total distance by 2, and then add that number to $-8{,}065$ *or subtract that number from* $6{,}893$.
5. Explain how to determine if the halfway point's elevation is above sea level or below sea level.
   Answers will vary. The halfway point will be at an elevation below sea level because Atacama Trench is farther below sea level than Mount Ojos del Salado is above sea level.

Consider showing video clips of Mount Ojos del Salado and Atacama Trench or mapping the location of Chile and Peru on a map to increase knowledge in context.

The following questions could facilitate connections for students struggling to discuss or explain what they see:

- *What's the total distance between the two elevations?*
- *What is half that distance? How far is each location from sea level?*
- *How can you use this information to determine whether the halfway point between the two locations is above or below sea level without performing the calculations?*

3.9.6 Your Turn!

Component Overview: *Students practice labeling a number line with the elevations of the captain and his fishing net. Students create an absolute value expression to determine the length of the rope. Next, students write and evaluate absolute value expressions to represent debt and temperature changes of chicken in a freezer.*

3.9.6 Your Turn!

1. The captain of a fishing vessel is standing on the deck at 23 feet above sea level. He holds a rope tied to a fishing net below him that is underwater at a depth of 38 feet.
   a. Draw a model of this situation using the number line.
   Answer shown on number line.
   b. Write a mathematical sentence using absolute value to determine the length of the rope.
   $|23| + |-38| = 61$

2. Aryan owes $\$325.46$ for recent purchases he made using his credit card and he owes his parents another $\$187.27$.

   a. Write a mathematical statement using absolute value that can be used to determine Aryan's total debt.
   $|-325.46| + |-187.27| = 512.73$
   b. What is the value of Aryan's total debt?
   $\$512.73$

3. When Brenda places a package of chicken in the freezer, the chicken is at a temperature of 41° Fahrenheit ($^{\circ}$F). While in the freezer, the temperature of the chicken changes $^{-}2^{\circ}$F every hour until it reaches the temperature of the freezer. What is the temperature of the chicken 3 hours after Brenda puts it in the freezer?
   $|41| - 3|^{-}2| = 35^{\circ}\text{F}$

Considerations to differentiate based on student need:

- *Facilitate small group conversations about what key points need to be plotted on the number line.*
- *Review the number line from 3.9.3 Guided Instruction.*
- *Prompt students to create an open number line with only the locations of 0, 23, and -38. Make sure students plot -38 farther from 0 than 23.*

- *How is a fishing net used?*
- *How is a credit card used?*
- *What happens to the temperature of meat when you place it in the freezer?*

3.9.7 Wrap-Up: Reflection

Component Overview: *Students reflect on their mastery of the unit learning targets related to absolute value. Use the data from this formative assessment to determine which students need remediation before the next lesson.*

3.9.7 Wrap-Up: Reflection

1. Check all of the statements below that describe your understanding and ability of working with absolute values.
   Answers will vary.

   - ☐ I can compare the absolute value of two numbers. I need support on how to compare the absolute value of two numbers.
   - ☐ I can perform operations that involve the absolute value of two numbers.
   - ☐ I need support to perform operations that involve absolute value.
   - ☐ I can interpret comparisons of absolute values in real-world contexts.
   - ☐ I need support interpreting comparisons of absolute values in real-world contexts.
   - ☐ I can perform operations that involve absolute value when solving real-world problems.
   - ☐ I need support to perform operations that involve absolute value when solving real-world problems.

Consider which lessons to review based on students' responses:

- *"I can compare the absolute value of two numbers." (Lesson 8)*
- *"I can perform operations that involve the absolute value of two numbers." (Lesson 8)*
- *"I can interpret comparisons of absolute values in real-world contexts." (Lesson 7)*
- *"I can perform operations that involve absolute value when solving real-world problems." (Lesson 9)*